

✓ Problem Set 2

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✓ 1 Hospital admission & quality of service

✓ Question 1

(a)

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import statsmodels.api as sm
import statsmodels.formula.api as smf
```

```
hdata = pd.read_csv('health_data.csv')
hdata.head()
```

	patient_id	hospital_id	admin_year	patient_died_dummy	startage	female_dummy
0	1	D	2003	0	81	0
1	2	H	2003	1	67	0
2	3	A	2003	0	54	0
3	4	E	2003	0	81	0
4	5	H	2003	0	69	0

```
model1 = smf.ols(formula = 'patient_died_dummy ~ hospital_id',data=hdata)
result1 = model1.fit()
print(result1.summary())
```

OLS Regression Results

Dep. Variable:	patient_died_dummy	R-squared:	0.042
Model:	OLS	Adj. R-squared:	0.042
Method:	Least Squares	F-statistic:	119.3
Date:	Mon, 10 Feb 2025	Prob (F-statistic):	1.75e-220
Time:	03:19:53	Log-Likelihood:	-7416.5
No. Observations:	24480	AIC:	1.485e+04
Df Residuals:	24470	BIC:	1.493e+04
Df Model:	9		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	0.0970	0.006	16.355	0.000	0.085	0.109
hospital_id[T.B]	0.0072	0.008	0.890	0.373	-0.009	0.023
hospital_id[T.C]	-0.0483	0.010	-4.776	0.000	-0.068	-0.028
hospital_id[T.D]	0.1882	0.008	23.188	0.000	0.172	0.204
hospital_id[T.E]	-0.0531	0.011	-4.771	0.000	-0.075	-0.031
hospital_id[T.F]	0.0003	0.009	0.030	0.976	-0.017	0.017
hospital_id[T.G]	0.0441	0.008	5.273	0.000	0.028	0.061
hospital_id[T.H]	0.0038	0.009	0.414	0.679	-0.014	0.022
hospital_id[T.I]	0.0318	0.009	3.480	0.001	0.014	0.050
hospital_id[T.J]	0.0112	0.011	1.028	0.304	-0.010	0.032

Omnibus:	9228.833	Durbin-Watson:	2.025
Prob(Omnibus):	0.000	Jarque-Bera (JB):	25954.225
Skew:	2.096	Prob(JB):	0.00
Kurtosis:	5.807	Cond. No.	10.0

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Intercept(0.097) represents the average mortality rate for the hospital A. The coefficient on the dummy for hospital D(0.1882) indicates the difference in the mortality rate between hospital D and hospital A. In other words, the mortality rate of hospital D is 0.1882 higher than the mortality rate of hospital A.

✓ (b)

```
result1.params['hospital_id[T.D]'] - result1.params['hospital_id[T.E]']
```

```
0.24139049162000606
```

The difference between the mortality rates at hospitals D and E is 0.24139049162000606. In other words, the mortality rate of hospital D is 0.24139049162000606 higher than the mortality rate of hospital E. This is because the `result1.params['hospital_id[T.D]']` represents the difference in the mortality rate between hospital D and hospital A and `result1.params['hospital_id[T.E]']` represents the difference in the mortality rate between hospital E and hospital A.

Question 2

(a)

```
hdata_AB = hdata[hdata['hospital_id'].isin(['A', 'B'])]
model2 = smf.ols(formula = 'patient_died_dummy ~ hospital_id', data=hdata_AB)
result2 = model2.fit()
print(result2.summary())
```

OLS Regression Results						
Dep. Variable:	patient_died_dummy	R-squared:				0.000
Model:	OLS	Adj. R-squared:				-0.000
Method:	Least Squares	F-statistic:				0.9377
Date:	Mon, 10 Feb 2025	Prob (F-statistic):				0.333
Time:	03:19:56	Log-Likelihood:				-1446.8
No. Observations:	6611	AIC:				2898.
Df Residuals:	6609	BIC:				2911.
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	0.0970	0.005	17.791	0.000	0.086	0.108
hospital_id[T.B]	0.0072	0.007	0.968	0.333	-0.007	0.022
Omnibus:	3377.258	Durbin-Watson:				2.031
Prob(Omnibus):	0.000	Jarque-Bera (JB):				14683.814
Skew:	2.650	Prob(JB):				0.00
Kurtosis:	8.022	Cond. No.				2.72

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Patients are not randomly assigned to hospitals. The patients go to the hospitals by their wishes. Some people may directly go to the hospital which is the nearest, some people may go to the famous hospital and some people may go to the hospital they have friends who work there, etc. There are many factors influence if the patient goes to which hospital. Also, the severity of illness, financial conditions, or other factors may influence which hospital a patient goes to. Furthermore, different hospital has different equipments, staff expertise, and histories, etc. This makes the direct comparisons misleading. Furthermore, some other variables like the health conditions of the patients may drive diverse hospital choice and mortality risk. Moreover, there are confounding factors such as if the patients in hospital B tend to have more severe conditions, then the mortality rate might be higher even if the hospital B provides same or better care and cure methods. Also, a hospital's reputation might also influence the choices of patients and some high-risk patients are more willing to go to the hospital with higher reputation. Thus, the difference in mortality rate of these two hospitals might be due to the differences in patient conditions rather than the choice of choosing these two hospitals. Thus, without randomization or proper controls, we cannot infer that moving a patient from Hospital A to B would change their probability of dying.

(b)

If we assume that patients with higher risks are more likely to go to the hospital B, then the difference in mortality between hospitals are overestimated. This is because the model is just included `hospital_id` as the independent variable. Then the effects such as the patients with higher risks and more severe conditions are more likely to go to the hospital B might increase the mortality rate of hospital B and this is attributed into the effects of the `hospital_id` in this model. Thus, in this case the difference in mortality between hospitals are overestimated. We might also consider that if hospital B is much better equipped, it might lower mortality despite receiving many sicker patients, leading to an underestimated effect. This bias makes direct comparisons unreliable without additional control variables.

(c)

```
model3 = smf.ols(formula = 'patient_died_dummy ~ hospital_id+startage+female_dummy+admin_year', data=hdata_AB)
result3 = model3.fit()
print(result3.summary())
```



OLS Regression Results

Dep. Variable:	patient_died_dummy	R-squared:	0.063
Model:	OLS	Adj. R-squared:	0.062
Method:	Least Squares	F-statistic:	110.7
Date:	Mon, 10 Feb 2025	Prob (F-statistic):	1.66e-91
Time:	03:45:12	Log-Likelihood:	-1232.7
No. Observations:	6611	AIC:	2475.
Df Residuals:	6606	BIC:	2509.
Df Model:	4		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-2.4995	5.161	-0.484	0.628	-12.616	7.617
hospital_id[T.B]	0.0114	0.007	1.581	0.114	-0.003	0.026
startage	-0.0010	0.000	-2.362	0.018	-0.002	-0.000
female_dummy	0.1838	0.009	21.018	0.000	0.167	0.201
admin_year	0.0013	0.003	0.507	0.612	-0.004	0.006

Omnibus:	3080.496	Durbin-Watson:	2.037
Prob(Omnibus):	0.000	Jarque-Bera (JB):	12181.549
Skew:	2.411	Prob(JB):	0.00
Kurtosis:	7.580	Cond. No.	2.89e+06

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
 [2] The condition number is large, 2.89e+06. This might indicate that there are strong multicollinearity or other numerical problems.

To move closer to a causal estimate, we need to control for some variables such as the startage, gender, and admin_year. We notice that the coefficient of hospital_id[T.B] becomes larger from 0.0072 to 0.0114. This indicates that the estimated difference in mortality rate between Hospital B and A increases after controlling for these three patient characteristics. This increase suggests that after accounting for some other variables, it is possible that Hospital B has higher mortality rate than Hospital A but it is not statistically significant since p-value (0.114) is larger than 0.05. The reasons why we include the specific set of variables are we hope to control for these patient characteristics variables and see the difference in the mortality rate between B and A more clearly in order to at least get closer to a causal estimate. We also observe that the p-value of startage (0.018) and female_dummy (0.000) is smaller than 0.05 and so these two variables are statistical significant. Thus, it is possible that these two influence the mortality rate for the hospitals. Also, older people might be more vulnerable to disease and it is possible that female might be more vulnerable to the disease. Furthermore, the global conditions which indicated by the admin_year might also influence the mortality rate. The medical technology, disease situations, and other global factors might affect the mortality rate. Thus, these three variables might be included in order to get closer to a causal estimate. By observing the p-values, the p-values for startage and female_dummy are smaller than 0.05 but admin_year is not. Thus, startage and female_dummy are statistically significant.

✓ 2 Demand estimation

Question 1

```
ddata = pd.read_csv('demand_data.csv')
ddata
```



	vendor_id	week	summer_dummy	price	sales
0	1	1	0	2.0	8788.7383
1	1	2	0	3.0	8937.9863
2	1	3	0	3.0	8740.1777
3	1	4	0	3.0	8757.1338
4	1	5	0	3.0	8739.6104
...	...	...	...	...	...
5195	100	48	0	2.0	8930.7900
5196	100	49	0	1.5	8583.6533
5197	100	50	0	1.5	8964.9170
5198	100	51	0	2.5	8811.0869
5199	100	52	0	1.5	8727.2656

5200 rows × 5 columns

```
ddata_vendor1 = ddata[ddata['vendor_id'] == 1]

model_sp = smf.ols(formula = 'sales ~ price',data=ddata_vendor1)
result_sp = model_sp.fit()
print(result_sp.summary())
```

OLS Regression Results

Dep. Variable:	sales	R-squared:	0.006
Model:	OLS	Adj. R-squared:	-0.013
Method:	Least Squares	F-statistic:	0.3250
Date:	Mon, 10 Feb 2025	Prob (F-statistic):	0.571
Time:	03:45:24	Log-Likelihood:	-360.33
No. Observations:	52	AIC:	724.7
Df Residuals:	50	BIC:	728.6
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	8983.8227	145.437	61.771	0.000	8691.704	9275.941
price	-31.2310	54.782	-0.570	0.571	-141.264	78.802

Omnibus:	3.319	Durbin-Watson:	1.572
Prob(Omnibus):	0.190	Jarque-Bera (JB):	2.367
Skew:	0.346	Prob(JB):	0.306
Kurtosis:	3.784	Cond. No.	12.5

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

```
model_sp_summer = smf.ols(formula = 'sales ~ price + summer_dummy',data=ddata_vendor1)
result_sp_summer = model_sp_summer.fit()
print(result_sp_summer.summary())
```

OLS Regression Results

Dep. Variable:	sales	R-squared:	0.318
Model:	OLS	Adj. R-squared:	0.290
Method:	Least Squares	F-statistic:	11.42
Date:	Mon, 03 Feb 2025	Prob (F-statistic):	8.49e-05
Time:	19:24:14	Log-Likelihood:	-350.56
No. Observations:	52	AIC:	707.1
Df Residuals:	49	BIC:	713.0
Df Model:	2		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	9177.5500	128.432	71.458	0.000	8919.455	9435.644
price	-141.1887	51.407	-2.746	0.008	-244.496	-37.882
summer_dummy	358.5012	75.790	4.730	0.000	206.195	510.807

Omnibus:	0.027	Durbin-Watson:	1.690
Prob(Omnibus):	0.986	Jarque-Bera (JB):	0.078
Skew:	0.039	Prob(JB):	0.962
Kurtosis:	2.828	Cond. No.	13.7

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

We observe that for the regression model of sales on price, the price coefficient is -31.2310. For the regression model of sales on price and summer_dummy, we find the price coefficient is -141.1887 and the coefficient for summer_dummy is 358.5012. Also, the p-value for the coefficient of price is 0.571 in the first model and it becomes 0.008 in the second model. This indicates that the price is statistically significant in the second model. After adding the summer dummy, the negative effect of price on sales becomes larger.

The omitted variable bias happens when a statistical model leaves out one or more relevant variables. The bias results in the model attributing the effect of the missing variables to those that were included. Based on the omitted variable bias formula, we know that the difference between the price coefficient when there is no summer dummy included and the price coefficient when there is summer dummy included is caused by the bias. When the summer dummy is not included, the effect of summer dummy is attributed to the price. In this case, the summer dummy coefficient is positive and it means the summer dummy has positive effects on sales. When there is no summer dummy in the model, this positive effect is included into the coefficient of price, making it become more positive than the true effect of the price on sales. If summer are related to both sales and price, omitting it from the model causes the price coefficient to be biased toward zero, which means the true effect of the price on sales is distorted. Once summer dummy is included, price's effect is properly accounted for, making the coefficient more negative and the p-value becomes much smaller.

Question 2

```
ddata_vendor2 = ddata[ddata['vendor_id'] == 2]

model_sp2 = smf.ols(formula = 'sales ~ price', data=ddata_vendor2)
result_sp2 = model_sp2.fit()
print(result_sp2.summary())
```

OLS Regression Results

Dep. Variable:	sales	R-squared:	0.133
Model:	OLS	Adj. R-squared:	0.116
Method:	Least Squares	F-statistic:	7.684
Date:	Mon, 10 Feb 2025	Prob (F-statistic):	0.00781
Time:	03:49:33	Log-Likelihood:	-359.10
No. Observations:	52	AIC:	722.2
Df Residuals:	50	BIC:	726.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	8411.1748	219.545	38.312	0.000	7970.205	8852.145
price	218.6028	78.863	2.772	0.008	60.202	377.004

Omnibus:	1.154	Durbin-Watson:	2.369
Prob(Omnibus):	0.562	Jarque-Bera (JB):	0.467
Skew:	0.114	Prob(JB):	0.792
Kurtosis:	3.404	Cond. No.	20.2

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

```
model_sp_summer2 = smf.ols(formula = 'sales ~ price + summer_dummy', data=ddata_vendor2)
result_sp_summer2 = model_sp_summer2.fit()
print(result_sp_summer2.summary())
```

OLS Regression Results

Dep. Variable:	sales	R-squared:	0.133
Model:	OLS	Adj. R-squared:	0.116
Method:	Least Squares	F-statistic:	7.684
Date:	Mon, 10 Feb 2025	Prob (F-statistic):	0.00781
Time:	03:49:36	Log-Likelihood:	-359.10
No. Observations:	52	AIC:	722.2
Df Residuals:	50	BIC:	726.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	2105.3159	29.848	70.534	0.000	2045.364	2165.268
price	2740.9463	10.951	250.283	0.000	2718.950	2762.943
summer_dummy	-2522.3436	75.986	-33.195	0.000	-2674.966	-2369.721

Omnibus:	1.154	Durbin-Watson:	2.369
Prob(Omnibus):	0.562	Jarque-Bera (JB):	0.467
Skew:	0.114	Prob(JB):	0.792
Kurtosis:	3.404	Cond. No.	1.33e+16

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The smallest eigenvalue is 2.61e-30. This might indicate that there are strong multicollinearity problems or that the design matrix is singular.

Based on the regression result, we find the smallest eigenvalue is 2.61e-30. This might indicate that there are strong multicollinearity problems or that the design matrix is singular. Also, we observe that the standard errors of the coefficients are large (75.986 for summer_dummy), which is another sign of multicollinearity.

The reasons that Price and Summer Dummy are highly correlated might be the vendor 2 only changes prices in summer and keeps price constant for other times, which would cause the price and summer_dummy are nearly collinear. Thus, the effects of price or the summer dummy on sales are hard to distinguish.

✓ Question 3

If one of the vendors did not systematically charge higher or lower prices in summer, the relationship between price and summer_dummy would be weak or nonexistent for this vendor. Thus, if we repeat the analysis we just did for vendors 1 and 2 for this vendor, without the summer dummy, we will observe that the coefficient for the price remain unbiased and the coefficient will capture the true relationship between sales and price without confounding from summer dummy. Also, its precision would be high since the standard error of price would

be low because price and summer dummy are not correlated. For the model with the summer dummy, the coefficient for price will remain stable because there is no collinearity between summer dummy and price. This means there is no large change for the coefficient of price when the summer dummy is added into the model. So the inclusion of summer dummy would not change the price coefficient a lot but the summer dummy might capture some variations in sales but it will not distort the coefficient for price. The precision will be high because the inclusion of summer dummy should not introduce multicollinearity since the summer dummy and price are not highly correlated. And so the standard error would be low and the precision is high.